

Lab Assignment 3

MTH 308: Numerical Analysis and Scientific Computing–I

Department of Mathematics and Statistics, IIT Kanpur

2026 Even Semester: January 30, 2026

This lab will have you implement and understand the effects of matrix ill-conditioning through the numerical examples provided below. Please write a single script file `Lab3LastnameRollNumber.m`. If you click on the **Publish** button in MATLAB, it will automatically create a folder named `html` containing a file named `Lab3LastnameRollNumber.html` and output figures. For example, if my roll number is 260102, my script file should be `Lab3Biswas260102.m`.

Each solution should:

- Be contained in a separate **code cell** (starting with `%%`) that includes the problem number.
- Run independently of other cells (include `clear` and `clc` where appropriate).
- Be adequately commented to explain your logic.

1. Write a function, which returns the condition number of the $n \times n$ Hilbert matrix, $A = [\frac{1}{i+j-1}]_{ij}$, and the error in the solution of a case of $Ax = b$, obtained via the backslash method, as explained below -

For the input n , generate the Hilbert matrix $A = \text{hilb}(n)$ and define a true solution vector $\mathbf{x}_{true} = [1, 1, \dots, 1]^T$. Construct the right-hand side vector $\mathbf{b} = A\mathbf{x}_{true}$ and solve the system $A\mathbf{x}_{comp} = \mathbf{b}$ using the MATLAB backslash operator (`\`). Finally, calculate the maximum absolute error $E_\infty = \|\mathbf{x}_{true} - \mathbf{x}_{comp}\|_\infty$. Your function should return both the maximum absolute error(e) and the condition number(k) of the matrix, A , and follow the format:

```
function [e,k] = Error_hilb(n)
- code to find e and k -
end
```

Also, write a function to:

- (a) Plot these errors against different values of n using a `loglog` scale.
- (b) Plot the condition number $\kappa(A)$ against different values of n .

Your function should follow the format:

```
function [] = plot_error_hilb(n1,n2)
- plot errors and condition numbers vs N on a single graph, where N = n1 : n2 -
end
```

Test the plot function for $N = 3 : 11$. Write your observations on how the error grows with n , and discuss how the condition number relates to the error computed in part (a). Then run the code for the range $N = 3 : 12$ and observe what changes. Does MATLAB display any warning messages?

2. Similarly, now let us see how the condition number varies with n for Vandermonde Matrices, and Tridiagonal Matrices.

- (a) **Vandermonde Matrix:** Create a grid of $n + 1$ points $x_i = i/n$ for $i = 0, 1, \dots, n$ (spanning $[0, 1]$). The $(n + 1) \times (n + 1)$ Vandermonde matrix is given by:

$$V = \begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix}$$

Write the following functions to compute the condition numbers of Vandermonde matrices and plot them against different values of n :

```
function k = cond_no_vand(n)
- generate the  $n \times n$  Vandermonde Matrix, using the definition above and return
its condition number ( $k$ )-
end
function [] = plot_cond_no_vand( $n_1, n_2$ )
- code to plot condition numbers vs  $N$ , where  $N = n_1 : n_2$  -
end
```

Test the plot function for $N = 4 : 17$.

(b) **Tridiagonal Matrix:** The tridiagonal matrix of size $n \times n$ is given by:

$$A_n = \begin{bmatrix} 2 & -1 & & & & \\ -1 & 2 & -1 & & & \\ & -1 & 2 & -1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & -1 & 2 & -1 \\ & & & & -1 & 2 \end{bmatrix}.$$

Write the following functions to compute the condition numbers of Tridiagonal matrices and plot them against different values of n :

```
function k = cond_no_tridiag(n)
- generate the  $n \times n$  Tridiagonal Matrix, using the definition above and return
its condition number ( $k$ )-
end
function [] = plot_cond_no_tridiag( $n_1, n_2$ )
- code to plot condition numbers vs  $N$ , where  $N = n_1 : n_2$  -
end
```

Test the plot function for $N = 5 : 200$.

Remark 1. Note that the MATLAB function `cond(A)` determines the condition number using the 2-norm by default. You may also specify other norms, such as `cond(A, 1)` for the 1-norm or `cond(A, inf)` for the ∞ -norm, to see if the choice of norm significantly changes your observations.